

# 00 — Table of Content / 总目录

Research Coding Handbook: guides 01–10, chapter 03's worked example, and supplementary Koopman learning materials.

科研编程手册总目录：01–10 主教程、第 03 章补充案例，以及 Koopman 学习资料。

Select an English title to open the English source, or a Chinese title to open the paired English–Chinese edition. Each chapter PDF has its own page numbers.

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## Chapters and worked examples / 章节与补充案例

| Chapter / topic<br>章节 / 主题                                                            | Contents<br>主要内容                                                                                                           |
|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>01</b> <a href="#">Inspect functions in Spyder</a><br>在 Spyder 中检查函数               | Breakpoints, stepping, variable types and shapes, local/global randomness.<br>断点、单步执行、变量类型与形状、局部及全局随机数。                    |
| <b>02</b> <a href="#">Insert and manage Markdown images</a><br>Markdown 插图与图片管理       | Relative paths, embedded figures, VS Code screenshots, display width.<br>相对路径、嵌入图片、VS Code 截图、显示宽度。                        |
| <b>03</b> <a href="#">Python data types and a profit equation</a><br>Python 数据类型与利润方程 | Built-in types, mutation, NumPy arrays, Case 1 calculations.<br>内置类型、可变性、NumPy 数组、Case 1 计算。                               |
| <b>03 · Example 1</b> <a href="#">MO-book production planning</a><br>MO-book 生产计划     | Equations to Python objects, LP variables, pandas, solver results and verification.<br>公式到 Python 对象、LP 变量、pandas、求解结果与验证。 |
| <b>04</b> <a href="#">Manual GitHub synchronization</a><br>GitHub 手动同步                | Remote checks, pull, stage, commit, push, verification and troubleshooting.<br>远程检查、拉取、暂存、提交、推送、结果核对与故障处理。                 |
| <b>05</b> <a href="#">Dunder methods and special names</a><br>双下划线方法与特殊名称             | Special methods, a profit container, entry-point guards, underscore patterns.<br>特殊方法、利润容器、入口判断、下划线命名形式。                   |
| <b>06</b> <a href="#">Create a GitHub organization</a><br>创建 GitHub 组织                | Account and organization names, setup, repository ownership, verification.<br>账号与组织名称、创建步骤、仓库归属、结果核对。                      |
| <b>07</b> <a href="#">Conda environments on Windows</a><br>Windows 下的 conda 环境        | Installation checks, environments, VS Code interpreters, reproducible specifications.<br>安装检查、环境创建、VS Code 解释器、可复现环境清单。    |
| <b>08</b> <a href="#">Python classes and objects</a><br>Python 类与对象                   | Classes, instances, self, initialization, attributes, methods, inheritance and composition.<br>类、实例、self、初始化、属性、方法、继承与组合。  |
| <b>09</b> <a href="#">Codex skills</a><br>Codex 技能                                    | Skill structure, installation paths, invocation, maintenance and scientific plotting.<br>技能结构、安装路径、调用、维护与科研绘图。             |

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|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>10 Imports, annotations and parsing</b><br>导入、类型标注与解析 | Imports, future annotations, argparse, __annotations__, __name__, dunder names, entry points and complete code.<br>导入、延后求值的标注、argparse、__annotations__、__name__、双下划线、入口判断与完整代码。 |

**Supplementary reading: Koopman operator / 补充阅读: Koopman 算子**

| Learning period<br>学习阶段                        | File<br>资料                                       |
|------------------------------------------------|--------------------------------------------------|
| Day 1<br>第 1 天                                 | <a href="#">Learning materials / 学习资料 (DOCX)</a> |
| Days 2–7<br>第 2–7 天                            | <a href="#">Learning materials / 学习资料 (DOCX)</a> |
| Days 8–14<br>第 8–14 天                          | <a href="#">Learning materials / 学习资料 (DOCX)</a> |
| Days 15–21<br>第 15–21 天                        | <a href="#">Learning materials / 学习资料 (DOCX)</a> |
| Days 22–30 and final project<br>第 22–30 天及最终项目 | <a href="#">Learning materials / 学习资料 (DOCX)</a> |